

Oil & Gas Seep Rates

① Oil-Bearing Rocks of The Davenport & Point Reyes areas & Their Implications for Offset along the San Gregorio & Northern San Andreas Faults:

Richard G. Stanley & Paul G. Lillis

- In both areas, much of the oil is in oil-bearing ss dikes & sills that intrude siliceous fine-grained strata of the Upper Miocene Santa Cruz Mudstone.
- 23 oil samples were all derived from Miocene source rocks (Davenport - 6, Point Reyes - 9, Half-moon Bay & La Honda oil fields - 8)
- Significant Biodegradation
- Oil Migration and ss intrusion happened $\approx$ 9-7 Ma and then the Davenport & Point Reyes areas were separated by 115 ± 10 km of right-lateral displacement along the S.G. & S.A. faults.
- Many of the ss intrusions are oil bearing but many are also oil free. Includes dikes & sills with width of few mm to more than 5 m and length of tens of meters.
- The ss intrusions formed by injection of fluidized sands from the Santa Margarita ss upward into fractured fine-grained strata of the Santa Cruz Mudstone.
- ** Some of the ss intrusions apparently were hydrocarbon-bearing at the time of emplacement (Phillips 1990; Thompson & others 1999) **Early HC Generation**
- Some of the ss intrusions were emplaced at very shallow sediment levels and some reached the sea floor and formed extrusives.

1/04

② Natural Oil & Gas Seepage in the Coastal Areas of California? James M. Galloway 1-16-04

- **Goleta Point**: 2 Containment Structures $\approx$ 4 BCF from 1982 - 2003 (22 years), $182 \text{ mmcf/year} \approx$ **500 mcfpd**
- **Famous Seeps**: La Brea Tar Pits, Pitch Lake of Trinidad, Perpetual Fires of Baku Azerbaijan.

2003

③ Compositional Changes in Natural Gas Bubble Plumes: Observations from the Coal Oil Point marine hydrocarbon seep field 2003

Jordan F. Clark, Ira Seifer, Julie Washburn

- Recent sonar surveys of the South Elwood anticline seep field indicate that about $1.5 \times 10^5 \text{ m}^3/\text{day}$ of seep gas is emitted from the 3 km^2 of seafloor.
 $\approx 5250 \text{ mCFPD}$ $1.916 \text{ TCF}/1000 \text{ YRS}$ or $19.16 \text{ TCF}/100,000 \text{ YRS}$ or $191.6 \text{ TCF}/1,000,000 \text{ YRS}$

191.6 TCF / 100,000 YRS
1916 TCF / 1,000,000 YRS

④ Temporal Variation in Natural Methane Seep Rate Due to Tides, Coal Oil Point area, California

J. R. Boles + J. F. Clark 11-15-2001

- 2 Large Tents (30m by 30m) Together = $30\text{m} \times 61\text{m} = 1830 \text{ m}^2$
 $= 19700 \text{ ft}^2$ Captures $16,800 \text{ m}^3/\text{day} = 594 \text{ mCFPD}$
 $594 \text{ mCFPD} / 19700 \text{ ft}^2 = 30.15 \text{ ft}^3/\text{ft}^2$

- Rate varies with tides; high tide correlates to reduced flow and low tide with increased flow. Each meter increase in sea height results in 1.5 to 2.2 % decrease in flow.

⑤ Dissolved Hydrocarbon Flux From Natural Marine Seeps to the Southern California Bight

Jordan F. Clark, Libby Washburn 5-15-2000

- Coal Oil Point seep field emits both oil & gas from more than 250 seafloor vents and accounts for 1/2 of all natural seepage along the Northern margin of the Santa Barbara Channel.
- Seafloor tents initially collected $.4 \times 10^5 \text{ m}^3/\text{day} = 1.413 \text{ mmCFPD}$ (In 1983) by mid-1990's the rate had declined to $.14 \times 10^5 = 495 \text{ mCFPD}$
- Using acoustical methods Quigley (1997) and Hornifino (1999) estimated the total emission of natural gas to the atmosphere to be between 1×10^5 and $2 \times 10^5 \text{ m}^3/\text{day}$ or 3.332 to 7.064 mmCFPD .

1.29 to 2.58 TCF / 1000 Years

129 to 258 TCF / 100,000 Years

1290 to 2580 TCF / 1 mm Years

12,900 to 25,800 TCF / 10 mm Years

129,000 to 258,000 TCF / 100 mm Years

22-141 50 SHEETS
22-142 100 SHEETS
22-144 200 SHEETS



5/00

11/99

⑦ Decrease in natural marine hydrocarbon seepage near Coal Oil Point, California, associated with offshore oil production. (Nov 1999)

Derek C. Quigley + J. Scott Horroff

- Seepage is most intense at submarine fault conduits and at structural closures along anticline axes.

- Seep Tests Rates Initial = 1.050 mmlFPD

Peak = 1.50 mmlFPD because of addition of 1/1995

1995 = 0.40 " " would be 0.750 mmlFPD without p.p.o. failure

$$0.75/1.5 = 50\%$$

- Seepage rate is proportional to the ΔP based on Darcy's Law. B6 Platform Holly production could be the cause. Gas injection at Holly was stopped in 1977 and by 1983 the reservoir pressure fell below hydrostatic and by 1994 the pressure was down 35%. Seep test emissions began a steep decline in 1990.

- This lag of several years can be explained as due to the slow K of the Monterey Formation migration pathways.

- The seepage area around Platform Holly has decreased by more than 50% in 22 years. Seep test volumes declined by 50% from 1989 to 1994.

$$1000 \text{ mmlFPD} / 19700 \text{ ft}^2 = .05076 \text{ mmlFPD} / \text{ft}^2 = 50.76 \text{ ft}^3/\text{day} / \text{ft}^2$$

7/01

⑧ www.vencoline.com (State of Ca Website) (7-03-01)

- Calif. State Lands Comm. has charted more than 1200 natural seeps in the Santa Barbara Channel. More than 1/2 occur within 3 miles of Coal Oil Point.

- Institute for Coastal Studies @ UCSB estimated the oil seepage near Coal Oil Point is 6000 gallons (143 Bbls) / day or 2.3 million Gallons/YR = 54,761 Bbl/year = 54.761 million / 1000 YRS = 54.761 Billion Bbl / myA (million YRS) = 2.738 Trillion Bbl / 50 myA (tertiary)

$$\text{Plus } 5 \text{ mmlFPD} = 1.825 \text{ BLFPD} = 1.825 \text{ TCF} / 1000 \text{ YRS} = 1.825 \text{ TCF} / \text{myA} = 91,250 \text{ TCF} / 50 \text{ myA (tertiary)}$$

$$\text{GOR} = 5000 \text{ MCF} / 143 \text{ Bbl} = 35 \text{ MCF} / \text{Bbl}$$



1999

⑥ The world's most spectacular marine hydrocarbon seeps (Coal Oil Point, Santa Barbara Channel, California): QUANTIFICATION OF EMISSIONS (1999)

J. Scott Hornafius, Dick Quigley & Bruce P. Legendre

- 50 KHz sonar data was used to estimate natural hydrocarbon emission rates from the 18 km² marine seep field offshore from Coal Oil Point

- Quantitative estimates of seep rates (From Figure 2):

un contained	La Brea seep field	= $8.9 \times 10^4 \text{ m}^3/\text{day}$ or $3.1 \times 10^6 \text{ CF/day}$
	Area around seep tents	= 1.3×10^4 " or 4.7×10^5 "
contained	man-made seep tents	= 1.0×10^4 " or 3.5×10^5 "
Nov 1994 Total		$11.2 \times 10^4 \text{ m}^3/\text{day}$ or $3.92 \times 10^6 \text{ CFPD}$

- From Abstract a body of report including associated oil emission rate of $1.6 \times 10^4 \text{ L/day}$ (100 BOPD) The total estimated gas emission rate is $1.7 \pm 0.3 \times 10^5 \text{ m}^3/\text{day}$ or $6.0 \pm 1.0 \times 10^6 \text{ CFPD}$

Our estimated methane emission rate for the Coal Oil Point seeps ($80 \pm 12 \text{ t/day}$) is 4 times higher than previous estimates.

* $6.004 \text{ MMCFPD} = 2.2 \text{ BCF/year} = 2200 \text{ BCF} (2.2 \text{ TCF}) / 1000 \text{ years} = 2200 \text{ TCF / Million years} = 110,000 \text{ TCF / 50 million years (Tertiary)}$

- Comparison of 7/95 data with 7/73 data suggests an 80% reduction in seepage rates within 1 km of platform Holly and a 50% reduction within 2 km. This decrease may be due to the reduction in reservoir pressure related to oil production. A reduction in natural seepage rates due to production has also been reported in Texas, Venezuela & Iraq. (Hornitz 1972, Landes 1973)

① Vegetation & geochemical prospecting for petroleum: AAPG B 36 (1972), p 925-940: L. Hornitz

② Methane nature as an oil pollutant: AAPG B 57 (1973) 637-644 K.K. Landes

- Hydrocarbon seeps comprise the largest source of marine methane emissions: 8-65 Trillion Grams = 8-65 million Tons Per Year.

- Their results for Coal Oil Point seep field of $1600 \text{ gm/cm}^2/\text{yr}$ 4 times the amount Hornitz used in estimate above.

⑥a Tracking oil slicks from the coal oil pt. seep field: J. Leiser AAPG Annual Mtg 2003

- 18 sq km field emits 105 m^3 of gas & 100 BOPD.

22-141 50 SHEETS
22-142 100 SHEETS
22-144 200 SHEETS



Most of present landscape formed in last 25 MY or late Cenozoic Tertiary 50 MYR

- ⑨ Other areas where seeps occur on Continental Shelves
- A) Gulf of Mexico (MacDonald 1998)
 - B) Gulf of California (Simoneit et al 1990)
 - C) North Sea (Hovland et al 1993)

1977 ⑩ Pomona College

- 1977 State Report identified 2000 leaks along coast of Santa Barbara that year, with another 665 inferred seeps. These amount to 500 BOPD, the area is also rich in natural gas seeps.

$$500 \text{ BOPD} = 182,500 \text{ BO/YR} = 182.5 \text{ Million BO/1000 YR} = 182.5 \text{ Billion BO/100 MYR} = 9.125 \text{ Trillion BO/50 MYR} \quad (\text{Tertiary})$$

- ⑪ 2001 GSA Mts Paper 42-0 abstract: Methane seepage along faults in the Santa Barbara coastal area, CA: Geology & modern evidence.

Jones R. Boles

- Tertiary sediments of the Santa Barbara basin are a prolific hydrocarbon source and there is good evidence that the basin has been leaking hydrocarbons, including methane, to the surface for at least 500,000 years.
- Over a 10^5 year time scale, seepage has evolved from high fluids & methane gas up along basin flank faults to cold gas seepage along faults within the basin. Based on U-Th dates from calcite in faults.

1972

⑫ TR 08: California Offshore Oil & Gas Seeps

E. R. Wilkinson (1972)

- Earliest record was in 1776 noted by Padre Pedro Font near La Bolea. Also 1792 by Captain Cook's Navigator in the S. B. Channel.

- Areas
 - 1) Point Conception (west of Coal Oil Point)
 - Magnitude may be as large as Coal Oil Pt
 - 1st documented by Seal Divers in the 1950's
 - # of separate vents
 - Sheet of Tar covering at least $1/4 \text{ mi}^2$ & forms a 10 to 12 foot seep, one of 3 localities in So. Calif. Pt. Conception, Coal Oil Pt & Carpinteria
 - From fractures in dipping Monterey Shale

2) Coal Oil Point (World's most publicized)

- Area of many seeps grouped into 3 main areas: Coal Oil Pt, La Bolea & Isla Vista



- 3) Platform "A" Seep (Caused by a blowout)
 - At least 1 natural seep in the area from a shallow oil sand in the Repetto formation.
- 4) Redondo Beach (Most active seeps south of Coal Oil Pt)
 - At least 6 separate seeps along the NW trending Pico Verde fault for about 12 miles. Redondo, Manhattan & Venice seeps included.
 - Estimates of flow based on slick size is $\approx 12-15$ BOPD per seep & undetermined gas. Constant stream of gas bubbles great attraction to fish.
- 5) Malibu (Caused by Earthquake of 2-9-71)
 - Limited amount of short duration seeps

(13) Earth Observatory (Nasa.gov). Media Alerts Archive
Roger Mitchell 301-231-0660

- 2 Times am Exxon Valdez spill worth of oil seeps into the G.O.M. each year according to Roger Mitchell - Earth Satellite Corp. There are over 600 different areas where oil oozes from rocks under the G.O.M.
- Calculated by using a thickness of $1/100$ th of a mm for the oil films, area covered by surface slicks (from Satellite Data) and est. how long oil stays on surface before it is consumed by bacteria or churned up by waves.

$$10.9 \text{ Million Gallons} = 260,000 \text{ BO} \times 2 = 520,000 \text{ BOPD} (1425 \text{ BOPD})$$

$$520 \text{ million BO} / 1000 \text{ years} = 520 \text{ BBO} / \text{million years} =$$

$$5200 \text{ BBO or } 5.2 \text{ TBO} / 10 \text{ MYA} = \underline{26 \text{ TBO} / 50 \text{ MYA (Tertiary)}}$$

2003

(14) The Fruitland Formation CBM Hydrocarbon System:
A Step-by-Step Analysis: W.L. Rouse: AAPG 2003 Ann Mtg.
Abstract Only

- Basin Subsidence Continued through the end of Cretaceous until Oligocene 40 to 35 mybp, when basin began to take present form.
- FC generated gas which filled cleats and was enhanced by meteoric waters. Uplift around 35 mybp allowed additional desorption and along with gas migration from deeper sources decreased K_{rw} sufficiently to prevent further fresh water invasion.
- Present-day methane seeps & springs at the outcrop are manifestations that the CBM system is still active.
- Fractures which have opened since Rio Grande Rift began to form ≈ 30 mybp, today allow fluids from the FC & deeper formations to come to the surface. This process has been ongoing for hundreds of years (demonstrably) - and have probably been active since the late Miocene.

2003 ⑮ Fruitland Formation Coal-Bed-Gas Seeps, No. 55B, Colo.
James F. Jassett; Paper 1-8; Rocky Mtn 55th Ann Mtg 2003
Abstract only

- Pine River Gas Seeps - Gas production from F-C CBM wells down-dip is unrelated to seeps.

3/00

⑯ Coal Bed Methane Development In The No. 55B of Colorado
A Brief History & Environmental Observation: BLM
55 Field office 12/99, Rev. 3-15-2000

History of Natural Gas Seeps in the Northern 55B

- Historically documented naturally occurring gas seeps throughout the 55B existed prior to oil & gas drilling operations.
- Shallow water wells penetrating Fruitland & Menefee coalbeds around the Basin rim have historically produced methane gas.
- Known gas seeps include the Carbon Junction area where the Animas River crosses the Fruitland Formation. At this location methane & H₂S seeps were commonly recognized as early as the 1930's.
- Another well-known site of historic gas seepage is a topographic low in the Hogback Monocline between Valenciel Cyn. & Iron Springs Cyn. on the western rim of the 55B.
- Other areas of seepage existed at the NE edge of the 55B rim. Residents ignited escaping natural gas from water faucets, holes punctured in ice over streams, or brown soil seeps in entertaining pyrotechnic displays.
- 40 years ago a group of local youngsters who inadvertently cast a campfire ember into the Las Pinos River were duly impressed when the surface of the river ignited in a flash (Hoeker, 1999).
- As early as 1980-1985, new seeps not associated with Basin rim outcrops, but interior to the basin, appeared to be forming in pastures in the Animas River Valley South of Durango.



1994

(17) Methane, CO₂, O₂, & N₂ in Soil Gas Overlying Coal Beds of the Up. Incl. Frt. Form. in the SSB, La Plata Co., SW Colo. USGS OFR 94-795: James R. Herrington: 1994

- Higher rank coal such as Low-Volatile B. bituminous & Anthracite, generates $\geq 100 \text{ cm}^3$ of methane per gram of coal
- Another Report SSB = 230 Billion Tons of FCC $1 \text{ gm} = .0022046 \text{ pounds}$

$$230,000,000,000 \text{ Tons} \times \frac{2000 \text{ lbs}}{\text{Ton}} \times \frac{453.592 \text{ grams}}{\text{lb.}} \times \frac{100 \text{ cm}^3}{\text{gram}}$$

$$= 2.0865 \times 10^{19} \text{ cm}^3 \text{ of methane} \times \frac{1 \text{ in}^3}{(2.54)^3 \text{ cm}^3} \times \frac{1 \text{ ft}^3}{(12)^3 \text{ in}^3}$$

$$2.0865 \times 10^{18} \text{ cm}^3 / 2.8317 \times 10^4 \text{ cm}^3/\text{ft}^3 = 7.3684 \times 10^{14} \text{ ft}^3 = 737 \text{ TCF}$$

A Seep A, released bubbling gas at 5.3 liters/minute

$$= 7632 \text{ liters/day} \times .2641721 \text{ gals/lt} \times \frac{1 \text{ gal}}{475 \text{ gals}} = 48 \text{ BPD}$$

1 Liter = .0062898 Barrels (US Pet)

50 SHEETS
22-142
100 SHEETS
22-143
200 SHEETS
22-144



6/01

(18) USGS Workshop on a Global Assessment of Geologically Sourced Methane: Tom Lorenson: June 2001

- Main objective to account for missing geologically sourced methane in the current global inventory of sources of atmospheric methane. The ancient-carbon, or geologically sourced, methane sources identified to date account for about 15% of the total methane in the atmosphere, which is 1/2 of the 30% needed to balance the observed atmospheric methane budget.
- As a 1st approximation, the observed global rate of methane emission at the seabed was estimated to be about 50 Tg/YR.
- Methane emissions from terrestrial sources is believed to be less than from the oceans but knowledge is minimal. Previous estimates from the exploitation of coal & petroleum are 35 to 45 Tg/YR.

$$50 \text{ Tg/YR} = 50 \times 10^{12} \text{ g/m/YR} = 50 \times 10^{12} \text{ g/m/YR} = 50 \times 10^9 \text{ kg/YR}$$

$$\text{If } 5.8752 \text{ kg/MCF}; \text{ then } = 8.51 \times 10^9 \text{ MCF/YR} = 8.51 \text{ TCF/YR}$$

$$= 8510 \text{ TCF/Year} = 8,510,000 \text{ TCF/MYA} =$$

$$= 425,500,000 \text{ TCF/50MYA (Tertiary)}$$

$$= 4.255 \times 10^8 \text{ TCF/50myA}$$

9/99

① Emission Inventory Guidebook (9-1-99)

- Compilation of reports on gas seepages indicates global emissions of 8 to 65 Tg/YR.
- Emission factors: 1) highest overall seepage rate is given for the Santa Barbara Channel at 400 gm/YR/m^2 ; 2) more typical factor given for the GOM, North Carolina Coast, & Danish Kattegat, should be used 50 gm/YR/m^2 .

$8 \text{ to } 65 \text{ Tg/YR} = 1.36 \text{ to } 11.06 \text{ TCF/YR}$

14 to 11 TCF/10 YR
140 to 110 " " 100 YR
1400 to 11,000 " " 1000 YR
14000 to 110,000 " " 10000 YR
140000 to 1,100,000 " " 100000 YR
1.4 to 11.1 " " 10 MYR
14 to 111 " " 100 MYR
140 to 1110 " " 1000 MYR

② Oil In The Sea III: GESAMP 1993

- 600,000 tons/year (Wilson et al 1973, 1974)
- 200,000 tons/year (Kuenvolden & Harbaugh 1983)
- 200,000 tons/YR to 2,000,000 tons/YR: revised GOM & SBC data applied to Kuenvolden & Harbaugh method

$600,000 \text{ tons} = 180 \text{ million Gallons} / 42 = 4.286 \text{ million BO}$
 So 1 ton = 7.1433 BO/Ton

$4.286 \text{ million BO/YR} = 4.286 \text{ Billion BO (BBO) / 1000 YR}$
 $= 4286 \text{ BBO / myA} = 42860 \text{ BBO / 10 myA} =$
 $2,143,000 \text{ BBO or } 2143 \text{ TBO / 50 myA (Tertiary)}$

Range @ 200,000 tons To 2,000,000 Tons / YR = $1.429 \text{ to } 14.29 \text{ million BO/YR}$
 $= 714 \text{ to } 7140 \text{ TBO / 50 myA (Tertiary)}$

- GOM = 140,000 Tons/YR, SoCal = 20000 Tons/YR, Alaska = 400 Tons/YR
Total NA = $160,400 \text{ Tons/YR}$

②A Oil In The Sea: Inputs, Fate & Effects (1985)

- Johnson (1971) & Larkin (1973) Compiled by Wilson (1973, 1974)
- Computations of 190 reported Submarine seeps can be augmented by 4 new ones: 1) Scott Inlet, Canada 2) Buchanan Gulf, Canada 3) Australian North Coast 4) Laguna de Jamialan, Mexico.

22-141 50 SHEETS
22-142 100 SHEETS
22-144 200 SHEETS



Natural Marine
Oil Seepage
Science (5-24-74)
1/184, # 4139

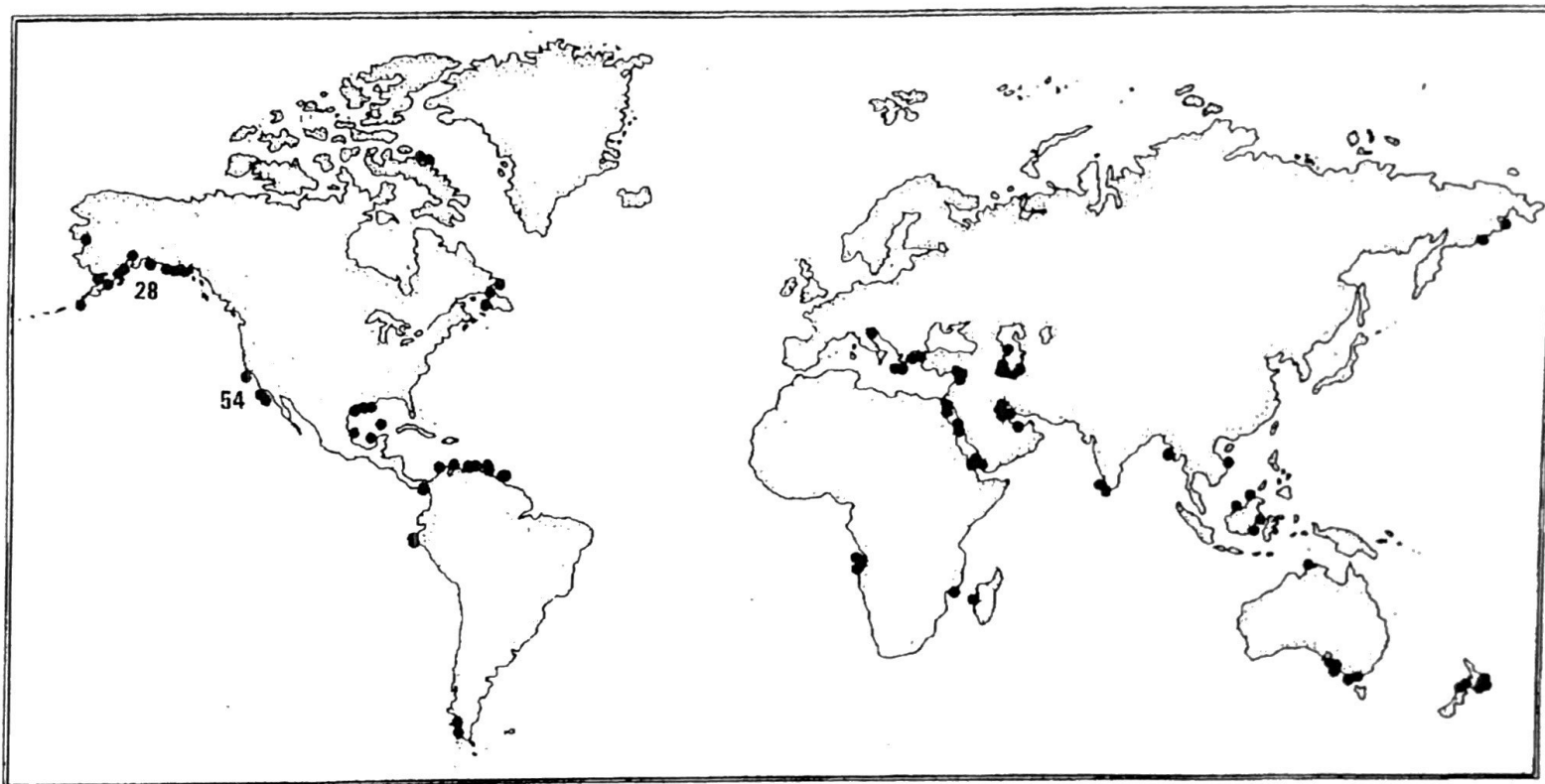


FIGURE 2-2 Location of reported marine seeps.

SOURCES: Wilson et al. (1973, 1974), Levy (1978), Levy and Ehrhardt (1981), McKirdy and Horvath (1976), Geyer and Giammona (1980), and Harvey et al. (1979).

Direct input of PHC from natural sources est. @ 0.025 - 2.5 million metric tons/yr
Best Est = 0.25 mTAC or 250,000 tons/yr

20A

TABLE 2-2 Petroleum Resource Estimates

Source	Amount		Reference
	Millions of Metric Tons	Billions of Barrels	
Offshore oil "tar" sands	30,000	200	Weeks (1965)
Total offshore petroleum resources	350,000	2,500	Weeks (1965)
Total petroleum resources in place	1,000,000	7,200	Hunt (1972)
Proven oil reserves and potential resources offshore	14,000	100	Albers et al. (1973)
Total proven reserves and potential resources onshore and offshore	86,000	630	Albers et al. (1973)
World exploitable oil (discovered 163,000 mt; undiscovered 141,000 mt)	304,000	2,200	Halbouty and Moody (1980)
Large "tar" deposits	320,000	2,100	Demaision (1977)

Geological Implications of Seepage Rates

A comparison of estimated seepage rates with the amount available for seepage can be used to assess the maximum geologic time during which seepages could be sustained.

Table 2-2 lists petroleum resource estimates for several categories of petroleum and illustrates the wide range of resource estimates that have been calculated. Table 2-3 illustrates the comparison. The wide range of assumed seepage rates includes the estimates of Wilson et al. (1973, 1974) but extends downward to 0.02 mta and upward to 10 mta. At the low end, 10,000 mt is near the value of 14,000 mt of total proven reserves and potential resources offshore as estimated by Albers et al. (1973). At the other end of the scale, Wilson et al. (1973, 1974) accepted an estimate of 300,000 mt in place. Because this amount may represent only 1% of the petroleum mobilized from source beds, they assumed that the amount available for seepage may be as much as 30,000,000 mt. This scale available for seepage has been augmented to 100,000,000 mt to attempt to include unknowns with regard to the amount of petroleum that would have been available for seepage during geologic time and will become available in the future during the lifetimes of the seepage.

Table 2-3 shows that to maintain petroleum seepage for a span of geologic time of at least 50 million years (most of the Tertiary period) requires that seepage rates be equal to or less than 2 mta, and

,025

TABLE 2-3 Maximum Lifetimes (Million Years) of World Oil Deposits

Oil Available for Seepage (mt)	Assumed Seepage Rates (mta)							
	0.02	0.04	0.20	0.60	1.0	2.0	6.0	10
10,000	0.5	0.25	.05	0.02	0.02	0.005	0.002	0.001
30,000	1.5	0.75	.15	0.05	0.03	0.015	0.005	0.003
100,000	5	2.5	0.5	0.2	0.1	0.05	0.02	0.01
300,000	15	7.5	1.5	0.5	0.3	0.15	0.05	0.03
1,000,000	50	25	5	2	1	0.5	0.2	0.1
3,000,000	150	75	15	5	3	1.5	0.5	0.3
10,000,000	500	250	50	20	10	5	2	1
30,000,000	1,500	750	150	50	30	15	5	3
100,000,000	5,000	2,500	500	200	100	50	20	10

at the same time the amount available for seepage must be equal to or greater than 1,000,000 mt. If seepage is maintained for 500 million years (most of the Phanerozoic), then seepage rates must be equal to or less than 0.02 mta and the amount available for seepage must be equal to or greater than 10,000,000 mt. The petroleum seepage rate that best seems to accommodate the requirements of reasonable geologic time and reasonable assumptions concerning availability for seepage is 0.2 mta with an uncertainty both upward and downward of an order of magnitude. Thus, the conclusion is reached that the average rate of petroleum seepage over time ranges from 0.02 to 2.0 mta, with a best estimate of 0.2 mta. This value is lower by a factor of 3 than the best estimate of Wilson et al. (1973, 1974) of 0.6 mta.

Erosional Inputs of Petroleum

The amount of petroleum that enters the marine environment by erosional processes has not been estimated before. Previous work by Wilson et al. (1973, 1974) considered only the marine input from natural seeps. Any estimation of erosional input of petroleum into the oceans can only be approximate.

There are at least three places where the erosional input of petroleum into the oceans could be studied in detail. Weaver (1969) showed examples of petroleum seeps at the beach and in the sea cliffs facing the Santa Barbara Channel, where erosion presently is taking place. Giammona (1980) described the Laguna de Tamiahua area where there are onshore and offshore seeps. The Marine Pollution Subcommittee of the British National Committee on Oceanic Research (1980) identified the Dorset coast of southern England as another place where petroleum source rocks as well as petroleum-containing reservoir rocks are exposed. They suggested this area for the study of natural seeps and erosional processes affecting the distribution of petroleum in the marine environment.

Because no direct information is available on erosional inputs of petroleum into the oceans, an indirect approach must be taken. This

- Caspian Sea Area; Baku - elevated firs
- South America: 1) Venezuela & Trinidad: Guayana pitch lake, Quirequire seeps, Mere Grenke, Lake Maricao El Mere, LePuy
- 2) Colombia: Infantas & La Canea Pools
- 3) Peru: de Brea Tar Sands & seeps
- Mexico: Tuxtepec Region thousands of seeps
- Canada: World's largest fossil surface seeps - Athabasca

Notes from Abstracts of the AAPG/Hodberg Conference 9/11-19/2007

- ① Key Issue Remaining: migration mechanism capable of moving hydrocarbons in a relatively short time (weeks to years) vertically through thousands of meters of strata w/o observable faults or fractures.
Proposed mechanisms are:
 - 1) Diffusion: Most likely contributes to near-surface movement but cannot account for relatively rapid movements observed in some studies.
 - 2) Effusion: requires movement as immiscible fluid through water saturated pore spaces. would require porous pathway to overcome capillary forces.
 - 3) Advection with moving waters: requires significant vertical water movement.
 - 4) Colloidal Bubble Ascent: microbubbles move vertically via a micro-fracture network.
 - 5) Continuous Gas Phase Flow: in fractures where small gas bubbles migrate as a continuous non-wetting phase in water-rich rock through microfractures.
- ⑦ Some Paleoseeps are affiliated w/ ancient sedimentary serpentinites that mark dewatering of subjacent ophiolites, tectonosomes, or paleo-gas hydrates. Evidenced by concentric, cement-lined conduits, carbonate chimneys, subvertical pipes beneath carbonate mounds.
- ⑩ Northern GOM: for seep sites surveyed, average total flux of gas (stored as hydrate), extruded mud & domed - biogenic precipitates over $\approx$ last 3000 years may be expected to be
 $\approx 3.3 \times 10^{10} \text{ gms/year} = 3.3 \times 10^7 \text{ kg/year} = 3.638 \times 10^4 \text{ tons/year}$
 at $9.07184 \times 10^5 \text{ grams/ton}$
 If assume 5.8752 kg/Mcf then $= 5.62 \times 10^6 \text{ Mcf/year} = 5.62 \text{ BCF/year}$

2002

(21) Transfer of hydrocarbons from natural seeps to the water column + atmosphere: I.R. Macdonald
Geofluids (2002) 2, 95-107 180 = 15% tars

- Kuervolden & Harbaugh (1983): Best estimate of global inputs of 2.5×10^8 L/YR entering the world's oceans probably low & should revise upwards since
- Mitchell (1999) calculated that the GOM contributes oil at the rate of .4 to 1.1×10^8 L/YR. = 252,000 Bbl/YR
= 6,692,000 Bbl/YR
- The Northern GOM contains over 200 active seeps.

(22) Submarine Seeps: Are they a major source of open ocean oil pollution.

- According to Weeks: a max of 700 BBO is available offshore by primary recovery. Since 10% is NGLS, there's 630 BBO (100,000 x 10.5 mmbbls) is the max. amount of likely source for offshore oil seeps.
- It has been estimated that offshore seeps amount to 0.5% of annual production. This would imply that submarine seepage would deplete the ultimate reserves of free flowing oil in 20,000 years.
 $0.5\% \times 2200 \times 10^6 \text{ tons (1970)} = 11 \times 10^6 \text{ tons}$

* If this amount has been constant since the early Tertiary the amount of seepage would equal 2500 times the free flowing oil more than 1500 times the total oil in place before offshore production began.

Other Major Seep Locations

- (S1) Significance of Oil & Gas Seeps in World Oil Exploration
Walter K Link AAPG V36 #8 (8/52)
- (S2) Levorenson Chapter 2: The occurrence of petroleum
- (S3) The history of world shallow oil & gas: Chapter 1
Ruth Shelden Knowles
 - United States 1) Onshore California, 2) Pennsylvania, & 3) Rocky mts
 - Middle East 1) Iraq (Kirkuk), Iran - mostly oil.

= 1.572 mmbbl/YR
1.572 BBO/1000 YR
1.57 TBO/1 million YR
78.5 TBO/100 million YR

22-141 50 SHEETS
22-142 100 SHEETS
22-144 200 SHEETS



5-3

Table 1
Recognition of Oil and Gas Seeps

Clues	Oil	Gas
Direct		
Visual	• Oil springs (dark brown to colorless). • Oil lakes or puddles. • Oil drips from rocks. • Rocks or soil are stained brown or black. • Asphalt or tar residue and mounds, often mixed with earth or vegetation (in dry climates, many paraffinic oils evaporate without a trace). • Oil floats on water in puddles, creeks or lakes and forms iridescent films (these can also be formed by iron, but break into fragments when disturbed). • Mud volcanoes (if not solfataric). • Positive ultraviolet or chloroform test.	• Gas bubbles in water or oil. • (Eternal) fires (could be marsh gas or coal gas). • Mud volcanoes and mud springs.
Smell	• Oil smell on outcrop or in broken rock fragments.	• Gas smell (esp. the "rotten egg" smell caused by poisonous H_2S).
Noise	• "Plopping" noise from mud volcanoes or mud springs.	• Gas vents give off a hissing sound. • Gas can explode. • "Plopping" mud in mud volcanoes and springs.
Indirect		
Visual	• Tar lumps floating on water or washing up on beach (could be pollution). • Stunted vegetation, bare patches in forest. • Yellowish stain from sulphur; whitish/many-hued stain from bacteria. • Wild or domestic animals coated with oil or tar. • Burnt clays (porcellanites). • Ozarkite, manjak veins, etc. • Saltwater springs and licks (may be seep-connected).	• Stunted vegetation. • Yellowish, whitish or varicolor stain (see oil). • Burnt patches in forest (could be from lighting, etc.). • Salt springs or salt licks (sometimes). • Preferred watering places for animals ("carbonated" H_2O). • Accumulations of dead animals, especially in low spots due to poisoning from H_2S gas.
Other	• Native place names, eg. manjak,	• Native place names (see oil).

Table 2
Some Examples of Direct Production From Oil and Gas Seeps

Location	Production	Remarks	Reference
Natural flow production			
Worldwide range	1 to many 100s liters per day		Various
Tar Springs, California, USA	2-4 liters/day		Blumer, 1922
San Timoteo, Venezuela	50 liters/day		"
Canyon City, Colorado, USA	75 liters/day		"
Sulphur Mt., California, USA	150 liters/day		"
Coal Oil Point, California, USA	est. 8,000 to 140,000 liters/day	Offshore	Wilson, 1973
Santa Monica, California, USA	320-3,200 liters/day	Offshore	Mikolaj, 1972
Oil Creek, Penna., USA	10-20 liters/day		Mikolaj, 1972
Ahwaz, Iran	80-100 liters/day		Coomber, 1938
Maidan-i-Naph-tun, Iran	3,000 liters/day		Cunningham-Craig, 1920
Estimate of world-wide marine seepage	640,000 to 16-million liters/day "Best" estimate is 1.6-million liters/day		Wilson, 1973
Umiat, Alaska, USA	800 liters/day		Johnson, 1971
Shelikoff Strait Alaska, USA	100 liters/day	Offshore	Johnson, 1971

Table 2
continued

Location	Production	Remarks	Reference
La Brea Pitch Lake, Trinidad	100-150 tons/day	Estimated possible influx	Author
Seep production from hand-dug pits or shallow wells			
Ghana	800-1,200 liters/day	Wells	Khan, 1970
Barbados	700 BOPD and 1 MM cu ft gas per day	± 20 wells	Bull AAPG Annual Review, 1983
Dominican Republic	"several hundred BOPD"	Shallow wells Information not verifiable, not sustained	Vaughan, 1921
Guyana	140,000 cu ft gas per year	Water well	Kugler 1942
Burma	32,000 to 64,000 liters/day	± 500 pits	Tainsh 1950
Dagah Shabell Somali Republic	10 liters/day	Pit	Beeby-Thompson 1918
Production from seeps using moderate to sophisticated technology			
Athabasca, Canada	160,000 to 170,000 BOPD	Sophisticated high cost mining, upgrading and transportation	

Pitch Lake,
La Brea,
Trinidad

11.3 million
tons from
1893-1980

Mostly manual
mining, but
mechanized and
relatively sophis-
ticated upgrading
and shipping

Various

Selenizza,
Albania

11,000 tons
per year

1963 figure

Various

Val de Travers
Switzerland

15,000 to
50,000 tons/year

Production
ceased

Various

Teca and Cocorna,
Colombia

± 5,000 BOPD

Production
from 50 wells
on or near
seep (± 100 BOPD
on pump) and using
steam stimulation

*The Texaco
Star*, 1984

Austria

< 100,000 Bbls
cumulative

No production now

France

< 300,000 Bbls
cumulative

No production now

West Germany

300,000 Bbls
cumulative

No production now

Thailand
(Mae Fang)

± 100,000 Bbls
cumulative;
individual wells
avg. 10 BOPD appr.
End 1958 est. 350-
500 BOPD from
15 wells

Government
made major
efforts to
develop this
seep area between
1930-1960

Various

Japan
(Kanto)

Approx. 300-
million m³
gas p.a.

In shallow
wells

Various

⑫ 25 miles of Trinitite surveyed revealed strong seeps (above background levels) along 7 miles of OC.

⑬ Rate of Migration: MAX during mud volcano eruption 10^{-2} m/s
min microseepage 10^{-10} m/s

⑮ Onshore Calif? Pico Formation; Stallen gas identical to production gases from field 4.5 mi. SE which has produced 23 BCF of dry gas (Pico Form)
Measured seepage rates range upwards from about 1 mL/min to as much as 9 L/min. Note of gas.

⑯ Total net input to atmosphere at about 540 Tg (teragrams = 10^{12} g of methane) 1,000,000,000,000 Same as Trillion

- 1st Approx of global rate of seabed emissions estimated at about 50 Tg/YR

⑰ In 1975, the estimated rate of natural seepage of oil into the marine environment ranged widely from 200,000 to 6,000,000 mta with a best est. of 600,000 mta. Based on extensive geological considerations but used extrapolations from only a few highly suspect seeps. In 1985, estimates of individual oil seep rates had not changed. A revised estimate was calculated based on the amount of oil present that could seep over reasonable periods of geologic time. The new estimates ranged between 20,000 + 2,000,000 mta with a best est. of 200,000 mta.

- Current studies using new technologies calculated;

1) GOM = 150,000 mta

2) Calif = 20,000 "

3) offshore Alaska = 20,000 "

190,000 m Just Part of North America

⑱ Canada "The Hat pot": These seepages are inferred to represent the discharge of regional gas migration systems of 10's to 100's of kilometers.

⑲ Coal Oil Point: 100,000 cm of gas + 100 BOPD; 16 methane added to the atmosphere from exposed seeps could be double the amount now estimated to result from marine seeps to 20 - 100 trillion grams (Tg) per year.

- (24) We can potentially estimate the flux rate of hydrocarbons to the surface by using microbially based calculations to determine how much gas per unit of time is required to support the level of hydrocarbon degrading cultures present in soils.
Need whole article

- (32) Surface hydrocarbon seeps & soil geochemical anomalies have been shown to appear & disappear in relatively short times (weeks to months to years)

- 1) Horvitz (1969) documented the disappearance in < 20 years of a soil hydrocarbon anomaly above Hastings Field, Brazoria Co., TX.
- 2) Sivakorvan (1974) during Soc. Rec., a surface geochem anomaly increased in intensity over Hilbig Field in Bastrop Co., TX.
- 3) Coleman & others (1977) demonstrated leakage of gas from 1000-meter deep gas storage reservoirs into shallow waterwells within a 1 year period.
- 4) Jones & Burtrell (1996) summarized results for a H of studies of Hyd flux variations in natural seeps as well as underground storage reservoirs.
- 5) Hitzman & others (2001) & Rice & others (2001) - documented sign. changes in microseepage int. & distribution over existing fields in in < 1 year.
- 6) Jones & Burtrell (1996) - report soil gas concentrations above gas storage reservoirs respond within 2-4 days to press changes in the reservoir.

- Based on these & other studies, the rate of hydrocarbon migration & microseepage varies from less than 1 meter to tens of meters per day (App 1992, Mathews 1996, Klusman & Sneed 1996, Clayton & Daniels 1996, Brown 2000, Rice 2001).

- Not limited to microseepage

- 1) Whelan & others (1994) - documented new OGLs entering some of the Reservoirs in Eugene Island Block 330, in the GOM, off La. Significant Δ in oil composition in 3-10 years.
- 2) Nunn & Meul Boeck (in press) - 60m data, make strong case for 2-10 km of vertical migration of brine & assoc. hyd. on time scale of 1-2 million yrs.

- (34) Theories of hydrocarbon microseepage need to account for the robust nature of vertical migration, even in a case not expected to include significant vertical fracturing to the surface.

- (35) This suggests that alkanes present as free gas within the soil had a residence time of only a few days.

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need to order
several papers

AAPG HEDBERG CONFERENCE
"Near-Surface Hydrocarbon Migration: Mechanisms and Seepage Rates"
SEPTEMBER 16-19, 2001, VANCOUVER, BC, CANADA

The Dynamic Nature of Hydrocarbon Microseepage: An Overview

By

Dietmar Schumacher, Geo-Microbial Technologies, Inc.

Hydrocarbon microseepage rates and near-surface hydrocarbon concentrations can vary significantly over time. Surface hydrocarbon seeps and soil geochemical anomalies have been shown to appear and disappear in relatively short times, weeks to months to years. Horvitz (1969) documented the disappearance in less than 20 years of a soil hydrocarbon anomaly from above Hastings field in Brazoria County, Texas. Sivaborvon (1974) reported that a surface geochemical anomaly increased in intensity over Hibig field in Bastrop County, Texas, during repressurization for secondary recovery studies. Coleman and others (1977) demonstrated leakage of gas from 1000-meter deep gas storage reservoirs into shallow water wells within a one year period. Jones and Burtell (1996) summarized results for a number of studies of hydrocarbon flux variations in natural seeps as well as from underground storage reservoirs. Hitzman and others (2001) and Rice and others (2001) have both documented significant changes in microseepage intensity and distribution over existing fields in time intervals as short as one year.

These studies clearly demonstrate rapid variations in gas flux associated with pressure changes in gas storage caverns and other oil and gas reservoirs. For example, Jones and Burtell (1996) report that soil gas hydrocarbon concentrations above gas storage reservoirs respond within 2-4 days to pressure changes in the reservoir or storage cavern. Similarly, soil gas concentrations above an underground coal gasification reactor changed dramatically within days of significant changes in the status of the coal-burn. Based on these and other studies, the rate of hydrocarbon migration and microseepage varies from less than one meter per day to tens of meters per day (Arp, 1992; Matthews, 1996; Klusman and Saeed, 1996; Clayton and Dando, 1996; Brown, 2000; Rice and others, 2001).

Data supporting the dynamic nature of hydrocarbon migration is not limited to microseepage. Whelan and others (1994) have documented that new oil and gas are entering some of the reservoirs in Eugene Island Block 330 field in the Gulf of Mexico, offshore Louisiana. Significant changes in oil compositions were observed to be occurring over three to five to ten years. Furthermore, the oil compositions show that the new oil is migrating into the reservoir from deeper, hotter formations and is not just the result of localized production-related lateral migration. Nunn and Meulboeck (in press), also working with Gulf of Mexico data, make a strong case for 2-10 km of vertical migration of brine and associated hydrocarbons on time scales of 1-2 million years.

These observations coupled with those previously reported for near-surface microseepage, demonstrate quite clearly that hydrocarbon migration and hydrocarbon microseepage are dynamic processes, processes leading to measurable changes on a time scale of months to years.

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South Ellwood Field Data

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Reservoir Size = 1.2 BBO Expansion Area = 1.0 BBO

Cum (1966-2000) Oil = 60 MMBO EUR = 69.4 MMBO
Gas = 40 BCF EUR = 46 BCF

Monterey Only

Cum (1972-5/2006) Oil = 57.5 MMBO Peak Prod = 1984
Gas = 49.4 BBO 3.45 MMBO PY
WTR = 63.0 MM BW 2.95 BCF PY

Formation	BH Orig	PSI/Ft	BHT	GOR	Hans
Sisquoc	615	.455	99	—	10
Monterey	1600	.478	150	500	500
Rincon	2205	.441	190	346-1000	150
Vaqueros	3650	.619	208	3947-4800	60
Slope	3650	.608	208	3947-4800	150

① 1996 (AAPG ABST): Hornafius

- Substantial Reduction In Natural Seepage w/ 1 mile of PR Holly
- Since production from Monterey began in 1972
- Monterey is source for gas
- 60% decline in Seep Gas Captured by Tents (1 mi East)
- corresponds to 40% reduction in BHP under PR Holly
- 443 MM BO & ~ Same volume of water since 1972

② 12/2003 (AGU): Boles

- initial BHP 5% over hydrostatic. Currently 25% under HS.

③ 9-15-99 (JGR): Hornafius main (6)

- 18 km² marine seep field off Coal Oil Pt includes seep tent
- gas emission rate = $1.7 \times 10^5 \text{ m}^3/\text{day} = 6.004 \text{ MMCFD} = 2.192 \text{ BCF/year}$
- includes seep tent = $.19 \times 10^5$ " = 0.671 " = 0.25 "
- oil seep rate = $1.6 \times 10^4 \text{ L/day} = 100 \text{ BOPD} = 36500 \text{ BO/year}$

Tertiary 50 MAYR		Year	100Y	1000Y	10KY	100KY	1MM	10MM
1.1 M. Million TCF	Total Gas	2.192 BCF	219.2 BCF	2.192 TCF	21.92 TCF	219.2 TCF	2192 TCF	21920 TCF
125000 TCF	Tent Gas	6.25 BCF	25 BCF	250 BCF	2.5 TCF	25 TCF	250 TCF	2500 TCF
18.25 Trillion BO	Total Oil	36500 BO	3.65 MMBO	36.5 MMBO	365 MM	3.65 BBO	36.5 BBO	365 BBO

④ 11/99 (Geology): Quigley Main (7)

- Nearshore Seeps (Coal Oil Point): Oil seeped directly from the outcrop of the Monterey Formation exposed in the axis of the Coal Oil Point Anticline.
- Farther offshore seepage passes through the overlying Sisquoc Formation caprock and includes both oil & gas. The offshore gaseous seepage is controlled by the local geologic structure which trends west-northwest. Seepage is most intense at submarine fault conduits and at structural closures along anticline axis.



- Seep Tents were installed in 1983 $\approx$ 1 mile SE (Updip) from Platform Holly & the South Ellwood Field. The collection rate was approximately 1000 MCFPD. In 1986 Flaps installed on the tents increased the rate to $\approx$ 1600 MCFPD.
- Production from the Monterey began @ South Ellwood in 1972. Gas was re-injected in the field +: 1977. By 1983 pressures in the Monterey began to decline below hydrostatic. By 1994 the total pressure drop was $\approx$ 35%.
- The Seep rate declined rapidly in 1989, approximately 6400 RS. After pressure decline started, by 1994 the rate was only 600 + 500 MCFPD. Pressures in 1994 7.54 MPA / 11.9 MPA = % drop.
1.5 km from Platform under seep tents 9.65 / 11.9 MPA = % drop.
- Seepage area has declined by more than 50%.

⑤ 11-15-2001 (JGR): Boles & Clark MAIN ④

Measured Seep Rates

$$1982: 9 \times 10^5 \text{ m}^3/\text{mo} \times 35.32 = 317.88 \times 10^5 \text{ ft}^3/\text{mo}$$

$$= 31,788,000 \text{ CFPM} / 30.4 = 1045 \text{ MCFPD}$$

ADD Seep Tent 1986: $+ 3 \times 10^5 \text{ m}^3/\text{mo} \times 35.32 = 105.96 \times 10^5 \text{ ft}^3/\text{mo}$

SK: its $= +10,596,000 \text{ CFPM} / 30.4 = +349 \text{ MCFPD}$

Total = $1045 + 349 = 1394 \text{ MCFPD}$

Between 1986-1989: $1.4 \times 10^6 \text{ m}^3/\text{mo} \times 35.32 = 49.448 \times 10^6 \text{ ft}^3/\text{mo}$

Peak Rate $= 49,448,000 \text{ CFPM} / 30.4 = 1627 \text{ MCFPD}$

Between 1990-1994: $3 \times 10^5 \text{ m}^3/\text{mo} = 349 \text{ MCFPD}$

Drilled
T: 1999: $5 \times 10^5 \text{ m}^3/\text{mo} = 582 \text{ MCFPD}$

Ground Inc

